PRACTICAL- 4

Problem Definition

Add dynamic UI components such as collapsible FAQ, modal popup,image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

1.How are DOM elements selected and modified?

DOM elements are selected using methods like getElementById(), querySelector(), and querySelectorAll(). They can be modified using properties like innerHTML, textContent, style, and classList

2.Are event listeners attached correctly?

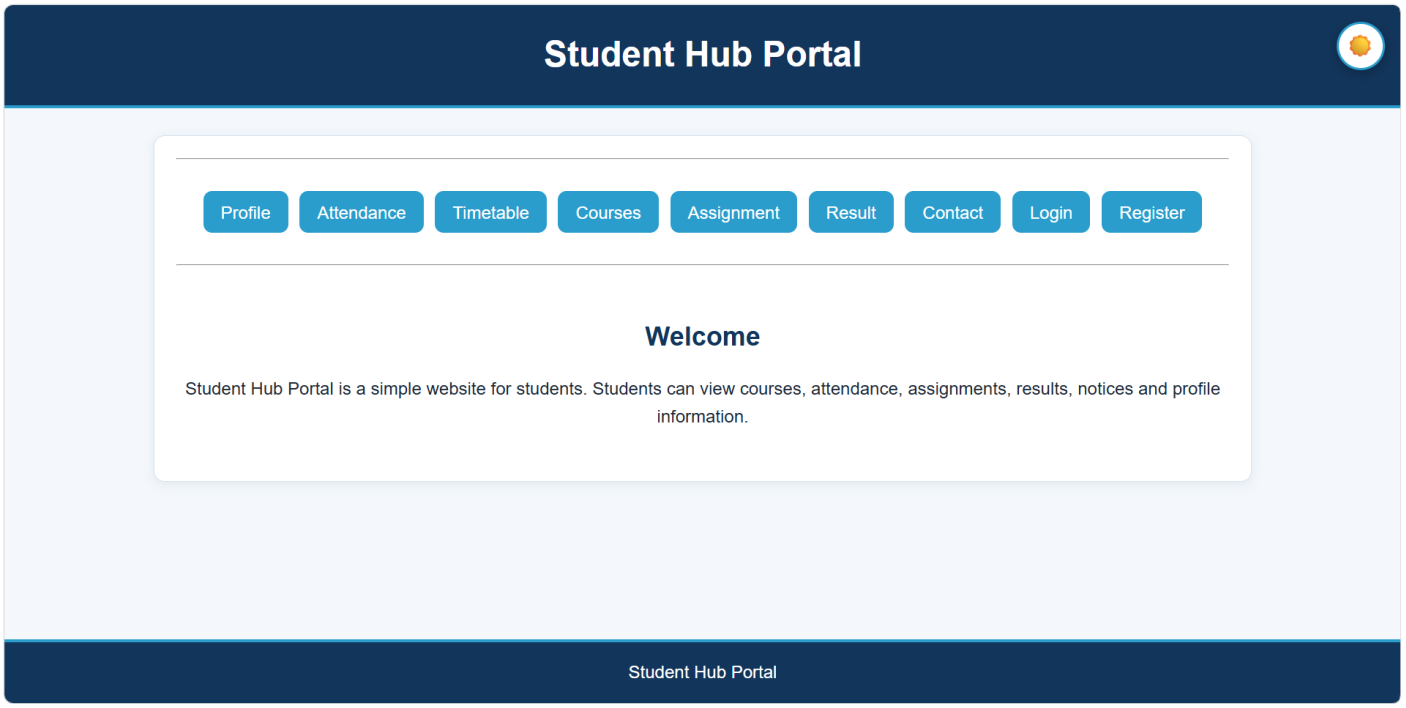
Yes. Event listeners are attached using addEventListener(), which allows JavaScript to respond to events such as click, submit, change, etc.

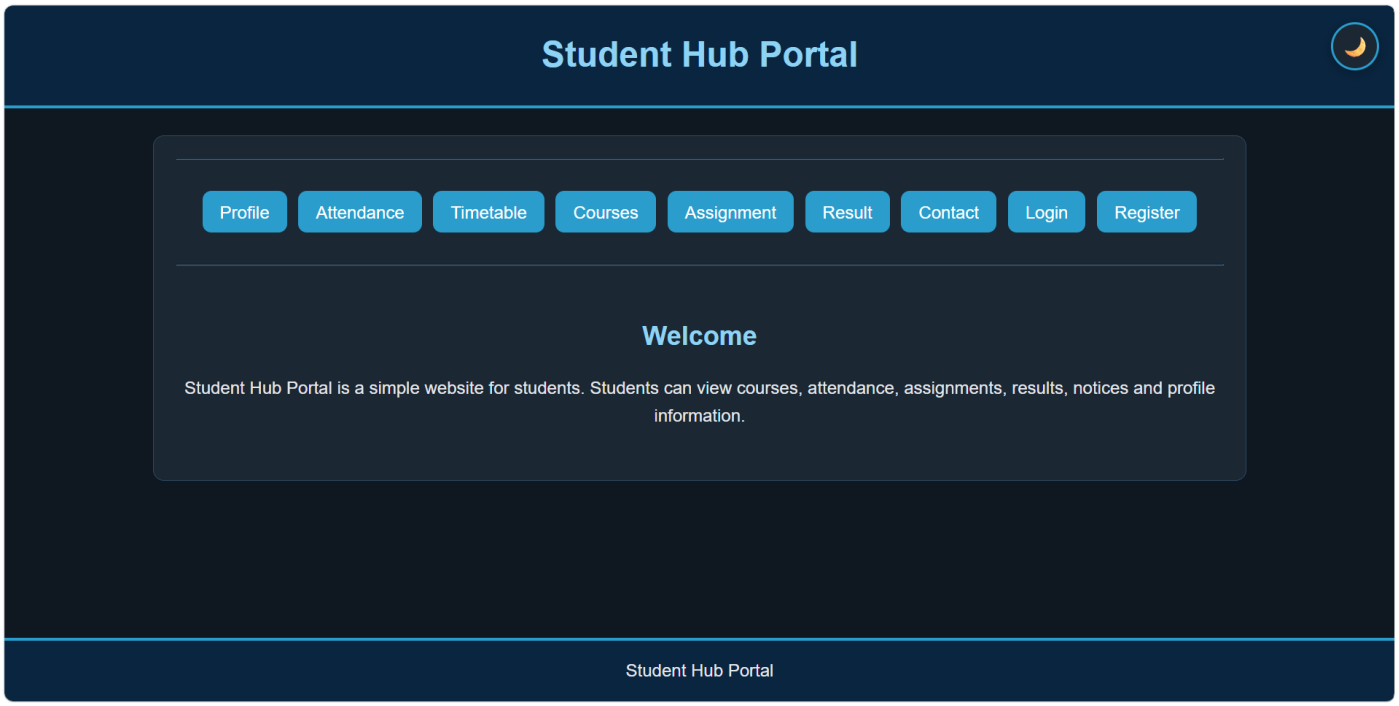
3.Is localStorage used for remembering theme choice?

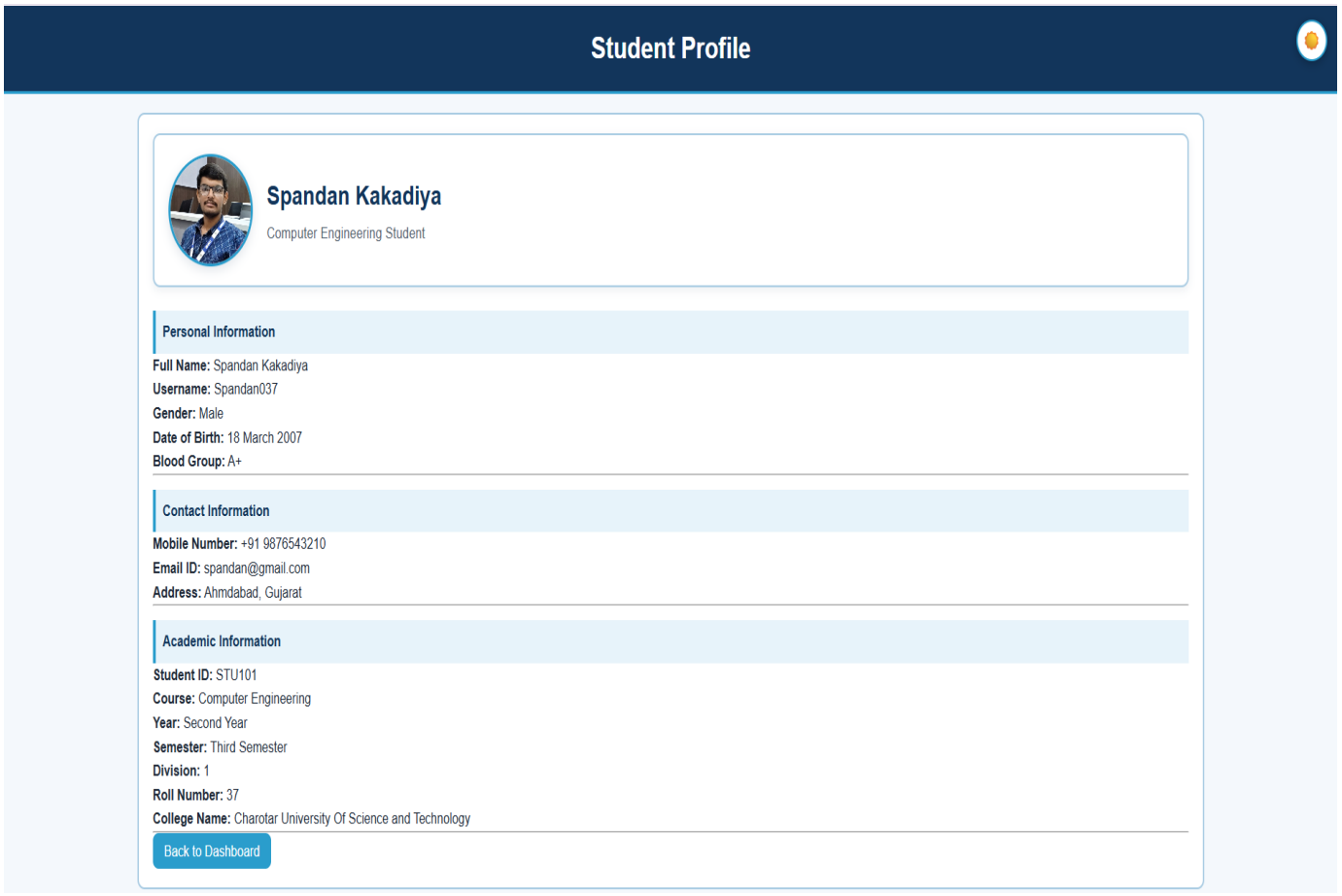
Yes. localStorage can store the user's selected light/dark theme so that the same theme is applied when the page is opened again.

4.Does interactivity improve usability without breaking accessibility? Yes. Features like menus, modals, notifications, and theme switching improve usability. They should also provide clear controls, readable text, keyboard access, and proper labels so accessibility is maintained.

SCREENSHOT :




Student Hub Portal
Profile
Attendance
Timetable
Courses
Assignment
Result
Contact
Login
Register
Welcome
Student Hub Portal is a simple website for students. Students can view courses, attendance, assignments, results, notices and profile information.
Student Hub Portal


Student Profile
Spandan Kakadiya
Computer Engineering Student
Personal Information
Full Name: Spandan Kakadiya
Username: Spandan037
Gender: Male
Date of Birth: 18 March 2007
Blood Group: A+
Contact Information
Mobile Number: +91 9876543210
Email ID: spandan@gmail.com
Address: Ahmdabad, Gujarat
Academic Information
Student ID: STU101
Course: Computer Engineering
Year: Second Year
Semester: Third Semester
Division: 1
Roll Number: 37
College Name: Charotar University Of Science and Technology
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Student Attendance Portal

Attendance Record

Fundamentals of Computer Networks	Total Classes	Classes Attended	Attendance %
Fundamentals of Data Structure and Algorithms	40	36	90%
Java Programming	45	40	89%
Discrete Mathematics	38	35	92%
Web Development Framework	42	39	93%
Creativity, Problem Solving and Innovation	42	39	93%

Check Attendance Clear Back to Dashboard

Time Table

Time	Monday	Tuesday	Webnesday	Thursday	Friday
9:10AM to 10:10AM	MSUD203	MSUD203	CEUC203	ITUC201	HSUV201
10:10AM to 11:10AM	MSUD203	CSUC201	ITUE201		
11:10AM to 12:10PM			Lunch Time		
12:10PM to 1:10PM	ITUE201	ITUC201	ITUE201	CSUC201	CEUC203
1:10PM to 2:10PM	MSUD201	CSUC201		ITUE201	HSUV201
2:10PM to 2:20PM			Break Time		
2:20PM to 3:20PM	CSUC201	CEUC203	CEUC203	ITUC201	HSUV201
3:20PM to 4:20PM			CEUC203	ITUE201	CEUC203

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Implementation :

- Created dynamic UI components using JavaScript.
- Implemented collapsible FAQ to show/hide answers.
- Added a modal popup that opens and closes on button clicks.
- Added a notification banner for user messages.
- Implemented a hamburger menu for navigation.
- Added a light/dark theme switcher using JavaScript.
- Used DOM manipulation and event handling to make the UI interactive.

Challenges :

- Handling click events for different UI components.
- Selecting and modifying DOM elements correctly.
- Managing show/hide functionality for FAQ, modal, and menu.
- Applying light and dark themes consistently.
- Making the components responsive and user-friendly.

GitHub Repository :

Link: https://github.com/25dce037-tech/Student-Hub-Portal.git

Conclusion :

The website successfully uses HTML, CSS, and JavaScript to create an interactive and user-friendly interface. DOM manipulation and event listeners provide dynamic functionality, while localStorage remembers the user's theme preference. Interactive components such as the menu, modal, notification, and slider improve usability while maintaining accessibility. Overall, the project demonstrates how JavaScript can make a web page more interactive, responsive, and user-friendly.